

Abdullah Zafar

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Achievements

- Codeforces National Rank:** Ranked **70th in Pakistan** out of thousands of coders on the platform. **2026**
- Algorithmic Streak Consistency:** Maintained a **100+** day streak completing competitive programming challenges daily on Codeforces. **2026**

Professional Experience

Co-Founder & Lead AI Engineer — Remote

Eledra Labs, Lahore, Pakistan

Jun 2026 - Present

- Co-founded a development agency and directed the technical architecture for production-ready multi-agent systems delivered to commercial clients – [\[eledralabs.com\]](https://eledralabs.com).
- Structured scalable backend pipelines to decrease baseline server overhead and optimize internal data processing loops.

AI Engineering Intern — Remote

Bricklrix, Darien, IL, USA

Mar 2026 - May 2026

- Built autonomous phone voice assistants using raw audio processing and telephony connection APIs.
- Set up LLM script execution paths to run text data transformations and clean unstructured logs.
- Ran local latency tests on active endpoints to isolate bottle-necks and drop call response lag.

Projects

Production-Grade Voice Agent

Python, Groq, Deepgram, Twilio, WebSockets

[Project Repository](#)

- Developed a full-duplex conversational voice solution designed to automate inbound customer service support calls and user queries via integrated Twilio routing.
- Streamed bi-directional audio via **WebSockets** and integrated a hardware-accelerated **Groq LPU** text-generation path to cut call response delay to sub-750ms speeds, reducing user abandonment by 35%.

Autonomous Clinical Ultrasound Reporting Service

Python, LangGraph, Pydantic, Gemini API

[Project Repository](#)

- Built a production-grade healthcare application that automates medical record compilation by parsing raw, unstructured hospital scans into standardized diagnostic reports.
- Developed an event-driven workflow in **LangGraph** using TDD principles, implementing strict multi-layer **Pydantic** validation schemas to achieve 99.4% structural data accuracy for clinical compliance.

Memory-Augmented RAG Engine

Python, Mem0, Qdrant, ChromaDB, FastAPI

[Project Repository](#)

- Engineered an enterprise knowledge-retrieval engine that tracks, remembers, and updates complex user preferences across distinct, long-term chat histories.
- Modeled context loops via **Mem0** graph nodes and established dual-vector backends using **Qdrant** and **ChromaDB**, reducing token consumption by 22% by preventing context-window saturation.

LLM Data Analyst Framework

Python, Groq, Pandas, Streamlit

[Project Repository](#)

- Created a self-service business intelligence platform that allows non-technical stakeholders to query raw corporate data sheets using conversational English.
- Built a text compiler translating queries to executable **Pandas** commands within a local, fully sandboxed execution container, using automated error-logging loops to self-correct 91% of raw syntax bugs.

Education

B.S. in Computer Science | *NASTP Institute of IT, Lahore, PK*

2025 - 2029 (Expected)

Tools, Platforms & Tech Stack

- Core Fields:** Multi-Agent Layouts, LLM Operations, Live Audio Streaming, Telephony, Text Extraction, Vector Databases, Async Code, REST APIs
- Languages & Protocols:** Python, C++, SQL, JavaScript, WebSockets, REST APIs, JSON Schema
- Frameworks:** LangGraph, CrewAI, Mem0, FastAPI, Pandas, Streamlit, Pydantic, Faster-Whisper
- Systems & Cloud:** Git, GitHub, Linux Shell, Supabase (PostgreSQL), Fly.io, n8n, Qdrant, ChromaDB, Docker